Gamma-Trace Transfer in the Bars Scalar Complex

The TK6B.6g local homotopy lemma and the remaining global gap

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Abstract

The scalar part of the proposed all-orders Bars III.B lift has been reduced to a homology-concentration problem for a five-term gamma-traceless Spin(8) resolution acting on the mass-graded HFT/DDF multiplicity module. The unresolved local question was how the projected-wedge differential should be combined with the Ramond-sector Clifford operator $Q^1 = \gamma \cdot \alpha_{-1}/\sqrt{2}$.

This note solves that local question exactly. The naive projected Clifford map is not a chain map, but its complete defect factors through the discarded gamma-trace sector. On the full spinor-form triangle the defect is repaired by a finite, explicit transfer involving the inverse trace Laplacian $(\Gamma G)^{-1}$. All relevant eigenvalues are positive and independent of mass level. What remains open is global: construct this filtered triangle on the physical pre-core module, prove compatibility with universal- supermultiplet division, and establish concentration of the resulting spectral sequence. No protected Spin(10) lift is used.

1 Why this operator problem appears

Let $V = 8_v$ be the transverse vector of Spin(8) and let S^+ and S^- be its two chiral spinors. The relative-Dirac analysis of the scalar Bars functional produces five gamma-traceless modules

$$U_p^\epsilon = \ker\Big(\Gamma_p^\epsilon : \Lambda^p V \otimes S^\epsilon \longrightarrow \Lambda^{p-1} V \otimes S^{\bar{\epsilon}}\Big), \qquad 0 \le p \le 4.$$

For one chirality their dimensions are

$$(\dim U_0, \dim U_1, \dim U_2, \dim U_3, \dim U_4) = (8, 56, 160, 224, 112).$$

Projected exterior multiplication gives an exact polynomial free resolution

$$0 \longrightarrow S_V(-4)U_0 \longrightarrow S_V(-3)U_1 \longrightarrow S_V(-2)U_2 \longrightarrow S_V(-1)U_3 \longrightarrow S_V U_4 \longrightarrow \mathcal{T} \longrightarrow 0,$$

where $S_V = \mathrm{Sym}(V)$ and $\mathcal{T}_n = [n, 0, 3, 0]_{D_4}$. Pairing this resolution with the cumulative core gives the exact scalar identity

$$(1 - q)\Phi = (1 + q)H_4 - (1 + q^2)B_\Gamma.$$

The remaining scalar theorem is a prefix dominance between B_Γ and H_4. Basile and Markou's orthosymplectic construction supplies the natural multiplicity-side operator

$$Q^1 = \frac{1}{\sqrt{2}}\,\gamma\cdot\alpha_{-1}, \qquad \{Q^1, Q^1\} = T^{11}.$$

Because Q^1 is a Lorentz scalar, it is not the adjacent map $U_p \to U_{p+1}$. At symbol level it is Clifford multiplication at fixed form degree. The question addressed here is how that fixed-degree operator interacts with the projected-wedge resolution.

2 The full spinor-form triangle

Put

$$A^\epsilon_p = \Lambda^p V \otimes S^\epsilon.$$

Besides gamma contraction Γ, define:

- G, exterior multiplication by the invariant gamma-valued one-form;
- E_v, exterior multiplication by $v \in V$; and
- $C_v = \gamma(v)$, Clifford multiplication on the spinor factor.

For $0 \le p \le 4$, gamma-traceless forms and gamma-wedge forms are complementary:

$$A^\epsilon_p = U^\epsilon_p \oplus G^{\bar\epsilon}_{p-1} A^{\bar\epsilon}_{p-1}.$$

Recursive substitution gives the triangular decomposition

$$A^\epsilon_p \simeq U^\epsilon_p \oplus U^{\bar\epsilon}_{p-1} \oplus U^\epsilon_{p-2} \oplus \cdots .$$

At the last degree used by the resolution this reads

$$A^\epsilon_4 \simeq 112 \oplus 224 \oplus 160 \oplus 56 \oplus 8.$$

This decomposition is exact over $\mathbb{Q}$ in the explicit split Clifford model. It is the finite triangle that must be retained when projected maps fail on its top row.

3 Four universal operator identities

Proposition 3.1 (Clifford-valued Cartan identities)**.** *For every $v \in V$ and every form degree, the ambient spinor-form operators satisfy*

$$\Gamma^\epsilon_{p+1} E^\epsilon_v + E^{\bar\epsilon}_v \Gamma^\epsilon_p = C^\epsilon_v,$$

$$\Gamma^{\bar\epsilon}_{p+1} G^\epsilon_p - G^{\bar\epsilon}_{p-1} \Gamma^\epsilon_p = (8-2p)\,\mathrm{id}_{A^\epsilon_p},$$

$$E^{\bar\epsilon}_v G^\epsilon_p + G^\epsilon_{p+1} E^\epsilon_v = 0,$$

and

$$C_v^{\bar\epsilon} G_p^{\epsilon} + G_p^{\bar\epsilon} C_v^{\epsilon} = 2E_v^{\epsilon}.$$

Proof. Write gamma contraction as the sum of form contraction followed by Clifford multiplication, and gamma wedge as the metric-dual sum of exterior and Clifford multiplication. The first and third identities follow from the ordinary contraction/exterior anticommutator and exterior nilpotence. The second and fourth use the Clifford relation $\gamma(x)\gamma(y) + \gamma(y)\gamma(x) = 2(x, y)$. The coefficient $8 - 2p$ is the difference between the vector dimension and twice the form degree. All signs above match the convention used by the rational matrix audit. □

The first identity is a finite algebraic counterpart of the relation between gamma contraction, exterior differentiation and the twisted Dirac operator in higher-spin Dirac complexes. See Bureš, arXiv:math/9901039. That analogy motivates the construction, but the proposition itself is an elementary Clifford calculation in the present modules.

4 The trace Laplacian and its inverse

Define

$$\Delta_p^{\epsilon} = \Gamma_{p+1}^{\bar\epsilon} G_p^{\epsilon} : A_p^{\epsilon} \longrightarrow A_p^{\epsilon}.$$

Proposition 4.1 (Trace spectrum)**.** *On the depth-j gamma-trace diagonal $U_{p-j}^{\epsilon_j}$ of A_p^{ϵ}, where $\epsilon_j = \epsilon$ for even j and $\epsilon_j = \bar\epsilon$ for odd j, the operator Δ_p^{ϵ} acts by the scalar*

$$\lambda_{p,j} = (j + 1)(8 - 2p + j), \qquad 0 \le j \le p.$$

Proof. On $U_p^{\epsilon} = \ker \Gamma_p^{\epsilon}$, the second operator identity gives $\Delta_p = 8 - 2p$. On an element $G_{p-1}y$ it gives the recursion

$$\Delta_p G_{p-1} y = G_{p-1}(\Delta_{p-1} + 8 - 2p)y.$$

Iterating down j trace diagonals sums the scalars $8 - 2p, 8 - 2(p - 1), \ldots, 8 - 2(p - j)$, which is the displayed formula. □

For the four outgoing maps $p = 0, 1, 2, 3$, the spectra are

p	spectrum of Δ_p
0	(8)
1	$(6, 14)$
2	$(4, 10, 18)$
3	$(2, 6, 12, 20)$

Every eigenvalue needed by the resolution is positive. If I_p and P_p denote inclusion and projection for U_p, and H_{p+1} is the coordinate along the gamma-wedge complement, then

$$H_{p+1} = (\Gamma_{p+1} G_p)^{-1} \Gamma_{p+1}.$$

Restricting the first Cartan identity to $I_p U_p$, where $\Gamma_p I_p = 0$, gives the explicit leakage formula

$$\boxed{H_{p+1} E_v I_p = (\Gamma_{p+1} G_p)^{-1} C_v I_p.}$$

The leading eigenvalue becomes zero at $p = 4$, but the five-term resolution has no outgoing $U_4 \to U_5$ map. Thus the transfer never divides by the middle-degree zero mode.

5 Why the naive projected Clifford map fails

Define the projected maps, with a prime denoting opposite chirality, by

$$\sigma_p = P_{p+1} E_v I_p, \qquad q_p = P'_p C_v I_p.$$

The natural commutator defect is

$$D_p = q_{p+1} \sigma_p - \sigma'_p q_p.$$

Put $R_p = 1 - I_p P_p = G_{p-1} H_p$. Since C_v and E_v commute on the full ambient spaces, inserting $1 = IP + R$ gives

$$D_p = -P'_{p+1} C_v R_{p+1} E_v I_p + P'_{p+1} E_v R'_p C_v I_p.$$

Every term therefore passes through a discarded trace diagonal. The identity $E_v G + G E_v = 0$ kills the second term after projection. The identity $C_v G + G C_v = 2E_v$ and the leakage formula reduce the first term to

$$\boxed{D_p = -2P'_{p+1} E_v (\Gamma_{p+1} G_p)^{-1} C_v I_p.}$$

This is a complete local answer. It proves both that the naive top-row map fails and that no arbitrary correction needs to be guessed.

The exact ranks of D_p are

vector orbit	$(\operatorname{rank} D_0, \ldots, \operatorname{rank} D_3)$
non-null	$(8, 48, 112, 112)$
nonzero null	$(4, 24, 56, 56)$

The two source chiralities give the same ranks. The full triangular matrices satisfy the ambient mixed relation exactly; the displayed defect is precisely the sum of the paths which leave and return to the top row.

6 What this changes in the all-orders programme

The local map problem is solved, but the Bars scalar sign has not yet been proved. The next statement must concern the physical multiplicity module, not another finite-dimensional Clifford identity.

A plausible route is now sharply specified:

1. Construct the mode-one vector action, the orthosymplectic trace algebra and $C = \sqrt{2}Q^1$ on a physical pre-core module whose character descends to the HFT massive-primary series.
2. Tensor that module with the complete spinor-form triangle and filter by gamma-trace depth.
3. Prove that the trace ideal and universal-supermultiplet division preserve this filtration. The off-diagonal transfer maps are the explicit operators above.
4. Prove concentration of the resulting spectral sequence and identify its two relevant edge homologies with the primitive series Π_4 and Π_3.
5. Deduce the equivalent B_Γ–H_4 prefix dominance and hence the scalar Bars correction $a_N \geq 0$.

Relative-BGG and k-Dirac resolutions provide a useful proof template for formal exactness of a filtered double complex; see Salač, arXiv:1705.09469 and arXiv:1705.10168. Their parabolic data are different, so they do not establish the HFT quotient statement. Likewise the orthogonal–orthosymplectic Howe decomposition of the unreduced supersymmetric tensor algebra, as in Cheng and Zhang, arXiv:math/0206036, does not by itself prove compatibility with massive-supermultiplet division.

The important gain is finiteness: the transfer coefficients and trace-depth range do not grow with mass level. Consequently the remaining all-orders problem is an analytic filtration and homology statement, not a need to inspect progressively higher Bars levels.

7 Exact audit and scope

The accompanying rational audit verifies:

- the decompositions $A_p^\epsilon = U_p^\epsilon \oplus GA_{p-1}^{\bar{\epsilon}}$;
- all four operator identities on a basis of the eight-dimensional vector space, and hence for every vector by linearity;
- the stated trace spectra and complement formula;
- exact factorization of every projected defect through trace coordinates;
- the full triangular mixed relation at null and non-null orbit representatives; and
- the previously established five-term resolution and its positive cokernel.

The integrated HFT/DDF audit remains exact through degree 20 and explicitly reports that no protected Spin(10) baseline was imported. Degree-20 positivity is evidence for the cumulative core; it is not used as a proof of the still-open homology-concentration statement.